

Agenda of the Day

- Process selection
- Processes and technology

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Be part of the conversation shaping how sustainability is engineered for the future.

WORLD ENGINEERING DAY FOR SUSTAINABLE DEVELOPMENT

4 MARCH '26
09:30-12:30
FOUNDERS' HALL

PROGRAM

ARRIVALS | 09:30 - 10:00

WELCOME REMARKS | 10:00 - 11:30
Funda Yagci Acar -
Vice President for Academic Affairs

Keynote
Soner Kahraman - Executive Director of R&D,
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Engineering Sustainability at Scale: R&D for a
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Merih Angin
Engineering Sustainable Futures with AI:
Measuring, Monitoring, and Acting on
Sustainability Data

Dr. Beril Boyacoglu
Koc Holding - KWORKS Climate Tech
Accelerator Program

Bora Cakirgözü
Sustainability & Entrepreneurship Today

PANEL DISCUSSION | 11:30 - 12:30
Engineering Sustainability: Where Energy, Industry,
and Systems Converge

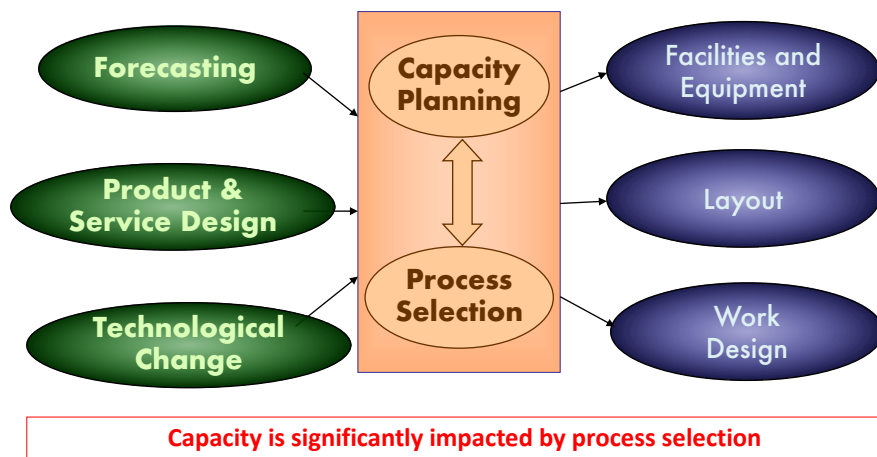
MODERATOR
Behice Pekilcan

PANELISTS
Cem Erkey
Ceyda Oğuz
Özlem Ünalı, Beiko
Soner Kahraman, Beiko

Join us for UNESCO World Engineering Day as we explore the role of engineering in
addressing today's sustainability challenges across research, industry, and education.

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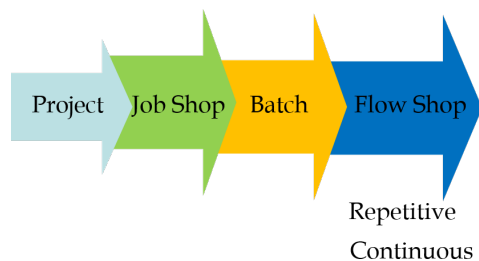
Process Selection and System Design



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Factors which influence process selection

- ❑ Product Variety
 - How much
- ❑ Flexibility
 - What degree
- ❑ Production Volume
 - Expected output



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Factors which influence process selection

□ Product Variety

- How much? → refers to different product designs or types that are produced in a plant.

➤ Flexibility

- What degree

➤ Production Volume

- Expected output

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Factors which influence process selection

□ Product Variety

- How much

□ Flexibility

- What degree

□ Production Volume

- Expected output: refers to the number of units of a given part or product produced annually by the plant

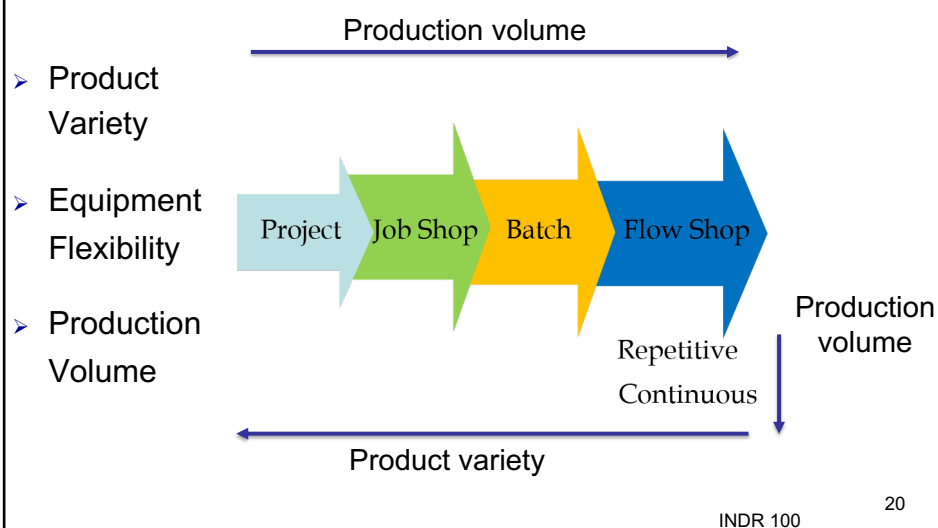
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Factors which influence process selection

- Product Variety
 - How much
- Flexibility
 - What degree of equipment flexibility?
- Production Volume
 - Expected output

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Factors which influence process selection



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There are three main types of production processes to choose from

- ❑ **Job shop production**, where items are made **individually**, and each item is finished before the next one is started. Designer dresses are made using the job production method.
- ❑ **Batch production**, where groups of items are made **together**. Each batch is finished before starting the next block of goods. For example, a baker first produces a batch of 50 white loaves. Only after they are completed will he or she start baking 50 loaves of brown bread.
- ❑ **Flow shop production**, where **identical, standardized items** are produced on an **assembly line**. Most cars are mass-produced in large factories using conveyor belts and expensive machinery such as robot arms. Workers have specialized jobs, for instance, fitting wheels.

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Job Shop Process



- ❑ Small firms operating in the service sector, such as plumbers, use **job shop production** because each customer has individual needs. Niche manufacturers of items such as made-to-measure suits would also use job production because each item they make is different.

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Batch process



- ❑ **Batch production** is used to meet group orders. For example, a set of machines could be set up to make 500 size 12 dresses and then adjusted to make 600 size 14 dresses. Two batches have been made.
- ❑ Or in a bakery, batches will form due to baking in an oven.

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Flow Shop Process



- ❑ **Flow shop production** is used to mass produce everyday standardized (all the same) items such as soap powder and canned drinks.

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Process Types

- ❑ **Project:** A non-repetitive set of activities directed toward a unique goal within a limited time frame
 - Unique
 - Examples: Building a bridge, consulting
- ❑ **Job shop:** provides unit or lot production or service with changeable specifications, according to customer needs
 - Small scale
 - Examples: Machine shop, carpenter, dentist's office
- ❑ **Batch:** Produces many different products in groups (batches)
 - Low or moderate volume
 - Examples: Bakeries, movie theaters, classrooms

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Process Types (Cont.)

- ❑ **Flow shop:** produces in high volume, low variety, and can be in two forms
 - **Repetitive:** provides one or a few highly standardized products or services
 - High volumes of standardized goods or services
 - Examples: automobiles, computers, cafeteria, car wash
 - **Continuous:** produces highly uniform products or continuous services, often performed by machines
 - Very high volumes of non-discrete goods
 - Examples: refineries, chemical plant, flour, sugar, electricity supplying and the internet

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Chocolate factory

- ❑ Different technologies and processes
- ❑ A repetitive/continuous process:
 - <https://www.youtube.com/watch?v=XNXWr5tg6ho>
- ❑ A job shop process:
 - <https://www.youtube.com/watch?v=TJMMYPAUkM0>

INDR 100

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- ❑ **Economies of scale** lead to lower unit costs and prices. Not many small manufacturers can afford the investment needed to mass produce goods. They instead opt for either batch or job production.
- ❑ There is usually a **trade off between unit costs and meeting specific customer needs**. Flow shop production offers economies of scale and low costs for a one-size-fits-all product.

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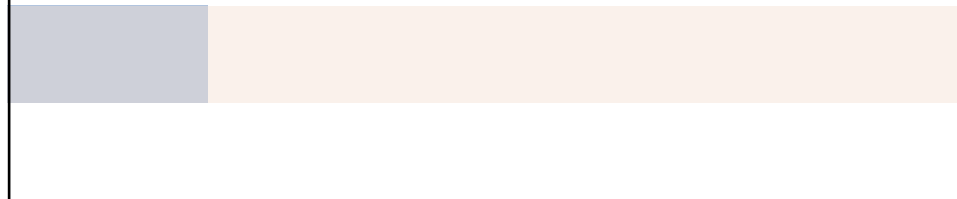
Process Choice Affects Activities /Functions

	Job Shop	Batch	Repetitive	Continuous	Projects
Cost estimation	Difficult	Somewhat routine	Routine	Routine	Simple to complex
Cost per unit	High	Moderate	Low	Low	Very high
Equipment used	General purpose	General purpose	Special purpose	Special purpose	Varied
Fixed costs	Low	Moderate	High	Very high	Very high
Variable costs	High	Moderate	Low	Very low	High
Labor skills	High	Moderate	Low	Low to high	Low to high
Marketing	Promote capacities	Promote capacities; Semi-standard goods/ services	Promote standardized goods/ services	Promote standardized goods/ services	Promote capacities
Scheduling	Complex	Moderately complex	Routine	Routine	Complex, subject to change
Work-in-process inventory	High	High	Low	Low	Varied

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Processes and technology



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Automation

- ❑ **Automation:** Machinery that has sensing and control devices that enables it to operate automatically
- ❑ Standardized goods and services
- ❑ Examples:
 - Goods: Automobile factories, semiconductors
 - Services: Package sorting, e-mail, on-line banking

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Automation Types

- ❑ **Fixed automation**
 - Specialized equipment for a fixed sequence of operations
- ❑ **Programmable automation**
 - Computer-aided design and manufacturing systems (CAD/CAM)
 - **Numerically controlled (NC) machines:** Machines that perform operations by following mathematical processing instructions.
 - **Robot:** A machine consisting of a mechanical arm, a power supply and a controller
- ❑ **Flexible automation**
 - Manufacturing cell
 - Flexible manufacturing systems
 - Computer-integrated manufacturing (CIM)

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Process Selection

- ❑ You are launching a startup making reusable bottles.
- ❑ You can buy an automated injection mold or hand-make them with molds and labor. Which process do you choose?

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Comparing Process Alternatives: Manual vs. Automated

Compare two or more process alternatives:

- one with high fixed cost & low variable cost (e.g., automation), and
- another with low fixed cost & high variable cost (e.g., manual assembly).

Mathematical Setup: For process i

$$C_i(Q) = F_i + v_i Q$$

F_i : fixed cost,

v_i : variable cost per unit,

Q : production quantity.

Find the break-even quantity between two processes i and j :

$$Q^* = \frac{F_j - F_i}{v_i - v_j}$$

Interpretation:

If $Q < Q^*$: choose the process with lower fixed cost (manual).

If $Q > Q^*$: choose the process with lower variable cost (automated).

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Comparing Process Alternatives: Manual vs. Automated

Example:

Process	Fixed Cost (₺)	Variable Cost (₺/unit)
Manual	5,000	25
Automated	20,000	10

$$Q^* = \frac{20,000 - 5,000}{25 - 10} = 1,000$$

So automation is better for volumes > 1000 units.

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addressing today's sustainability challenges across research, industry, and education.

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